

Simulation-Based Investigation of Variable Selection

Methods

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Introduction

Variable selection is frequently used in regression modeling to identify a subset of predictors that meaningfully explain variation in an outcome. Although common in applied research, selection procedures can introduce instability, inflate Type I error, and distort post selection inference, particularly in moderate sample sizes or when predictors are correlated. Despite these limitations, methods such as stepwise selection, AIC/BIC criteria, and penalized regression remain widely implemented. Therefore, understanding how these approaches behave under realistic conditions is important for investigators who rely on them.

This project investigates several common variable selection methods in linear regression using simulations and aims to evaluate how variable selection procedures perform when the true underlying model is known. Datasets will include 20 continuous predictors, 5 with (true) non-zero effects. Two sample sizes ($N = 250$ and $N = 500$) and several predictor correlation structures will be examined to represent moderate sample size studies with varying multicollinearity. Each method (backward selection based on p-values, AIC, BIC, LASSO, and elastic net) will be assessed on its ability to identify true predictors, avoid extraneous variables, and produce unbiased and well-calibrated estimates.

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Model Notation

Data were generated from the linear model $Y = \beta_0 + \sum_{j=1}^{20} \beta_j X_j + \epsilon$, with $\epsilon \sim N(0, 1)$. Predictors X_1 to X_5 had true coefficients ranging from $1/6$ to $5/6$ in increments of $1/6$, and the remaining 15 predictors had coefficients of zero. Predictor correlation followed an exchangeable structure with $\rho = 0, 0.35$, and 0.7 . The case $\rho = 0$ corresponds to independent predictors.

Methods

All analyses were be conducted in R v4.5.2 (2025-10-31 ucrt). Data was be generated using `gen_data` from the `hdrm` package. Five variable selection strategies will be applied to each simulation dataset:

1. Backward selection using p-values,
2. AIC-based selection,
3. BIC-based selection,
4. LASSO, and
5. Elastic net.

Backward selection using p-values removed variables sequentially until all remaining predictors had p-values below 0.05. AIC and BIC based backward selection used penalties of $k = 2$ and $k = \log(n)$, respectively. LASSO and elastic net models were fit using `glmnet` with standardized predictors with $\alpha = 1$ for LASSO, $\alpha = 0.5$ for elastic net. For each

penalized method, two tuning parameters will be evaluated: `lambda.min` (minimum cross-validated error) and `lambda.1se` (largest λ within one SE of the minimum) from 10-fold cross-validation. Predictors with non-zero coefficients at the chosen λ were considered selected. Because `glmnet` does not provide standard errors, each model was refit using `lm` to obtain coefficient estimates, confidence intervals, and p-values. Predictors not selected were treated as non-significant. This reflects how investigators typically interpret selected models and aligns with the project’s focus on post-selection inference.

Simultaion

Each scenario (combination of sample size and correlation level) was replicated 2,500 times, chosen to achieve a Monte Carlo standard error of approximately 0.01 for proportions near 0.5. For each replication, a dataset was generated, all selection methods were applied, and the selection model was refit. Across replications, performance was summarized using:

- True Positive Rates (TPR): proportion of simulations selecting each true predictor
- False Positive Rates (FPR): proportion of simulations selecting any noise predictor
- Bias: mean difference between estimated and true coefficients among selected true predictors
- Coverage: proportion of 95% confidence intervals containing the true coefficient
- Type I error: proportion of noise predictors declared significant
- Type II error: proportion of true predictors not declared significant

Results

Appendix

Code Directory: <https://github.com/eweible/BIOS6624.git>

File name: P4/Code/P4_Data_Analysis.Rmd